

Computing Magnetic Exchange Parameters from First Principles: The TB2J Approach

—with application to spin-lattice coupling and planewave projector interface—

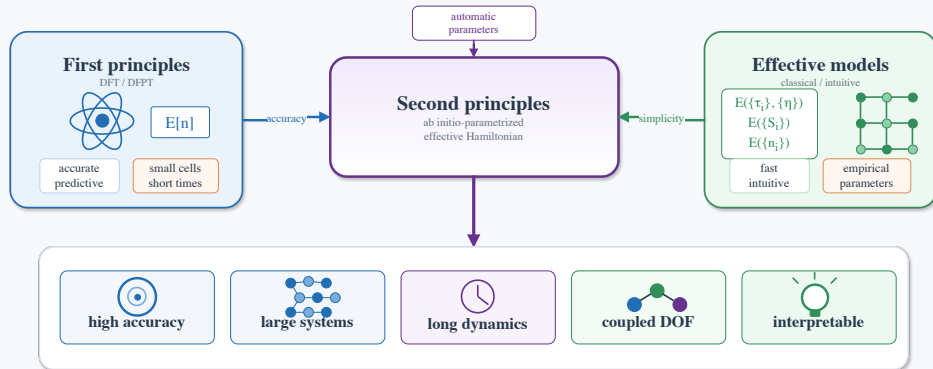
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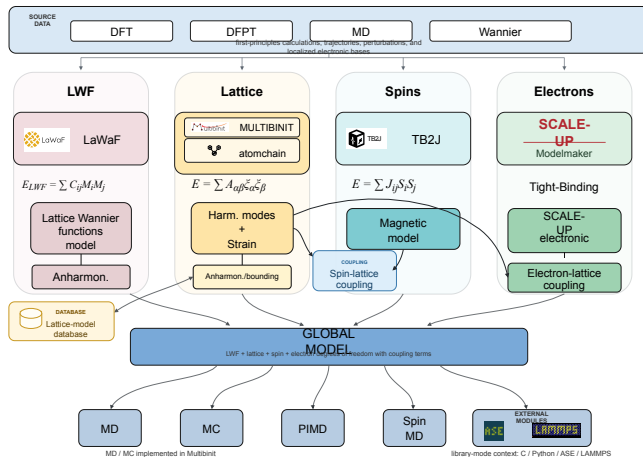
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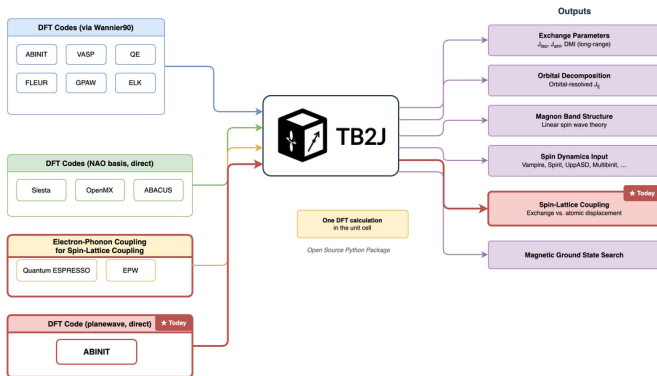
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@ABINIT Developer Meeting









TB2J compute the parameters of a generalized Heisenberg model from DFT ground state result.

$$E = - \sum_{i \neq j} [J_{ij}^{\text{iso}} \vec{S}_i \cdot \vec{S}_j + \vec{S}_i J_{ij}^{\text{ani}} \vec{S}_j + \vec{D}_{ij} \cdot (\vec{S}_i \times \vec{S}_j)]$$

using the **magnetic force theorem** and Green's function perturbation theory.
Requires only **one DFT calculation** in the unit cell.

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LOCAL SPIN DENSITY FUNCTIONAL APPROACH TO THE THEORY OF EXCHANGE INTERACTIONS IN FERROMAGNETIC METALS AND ALLOYS

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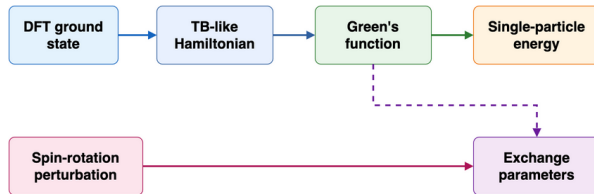
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Magnetic local force theorem

Change in total energy due to an **infinitesimal spin rotation** is approximated by the change in the sum of **single-particle energies**, without needing to recalculate the self-consistent electronic structure.

- Single-particle energy $\leftrightarrow$ band energy of TB-like Hamiltonian
 - ▶ Wannier functions
 - ▶ Pseudo-atomic orbitals
- Local spin-rotation $\leftrightarrow$ rotation of spin on localized basis
- Variation of single-particle energy $\leftrightarrow$ Green's function method



Refs: Korotin et al. 2015 Phys. Rev. B 91, 224405; Oroszlány et al. 2019 Phys. Rev. B 99, 224412; Durhuus et al. 2023 J. Phys.: Condens. Matter 35 105802

From tight-binding Hamiltonian

$$G(\vec{k}, \epsilon) = \left(\epsilon S(\vec{k}) - H(\vec{k}) \right)^{-1}$$
$$G(\vec{R}, \epsilon) = \frac{1}{N_k} \sum_{\vec{k}} G(\vec{k}, \epsilon) e^{-i\vec{k} \cdot \vec{R}}$$

Green's function perturbation expansion

$$\Delta E = -\frac{1}{\pi} \text{Im} \int_{-\infty}^{E_F} \text{Tr} [\delta H G + \delta H G \delta H G + \dots] d\epsilon$$

Collinear limit

Without SOC, anisotropic exchange and DMI vanish.

$$\Delta_i = H_i^{\uparrow} - H_i^{\downarrow}$$

LKAG formula

$$J_{ij} = -\frac{1}{4\pi} \int_{-\infty}^{E_F} \text{Im Tr} \left[\Delta_i G_{ij}^{\uparrow} \Delta_j G_{ji}^{\downarrow} \right] d\epsilon$$

Direct ABINIT-TB2J interface

Pseudo-atomic orbital projectors from planewave DFT

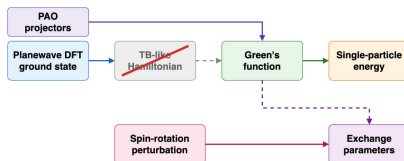
Planewave DFT to TB2J

For planewave codes, TB2J requires a mapping of the Hamiltonian to a tight-binding form, often through Wannierization.

Drawbacks of Wannier functions

- Not easily automatized.
- Disentanglement can be difficult.
- Symmetry is often not preserved.
- Long tails can extend to neighboring atoms.

Do we really need the tight-binding Hamiltonian?



Projector Green's function

$$G_{ij}^{\sigma}(\vec{k}, E) = \sum_n \frac{\langle \phi_i | \psi_{nk\sigma} \rangle \langle \psi_{nk\sigma} | \phi_j \rangle}{E + E_F - \epsilon_{nk\sigma}}$$

exchange

$$J_{ij} = -\frac{1}{4\pi} \int^{E_F} \text{Im Tr} \left[\Delta_i G_{ij}^{\downarrow}(E) \Delta_j G_{ji}^{\uparrow}(E) \right] dE$$

Onsite spin splitting Δ_i Two contributions:

- XC: $\Delta_i^{\text{XC}} = \langle \phi_i | B^{\text{XC}} | \phi_i \rangle$,
 $B^{\text{XC}} = V^{\text{XC},\uparrow} - V^{\text{XC},\downarrow}$
- DFT+U: $\Delta_i^{\text{+U}} = V^{\text{+U},\uparrow} - V^{\text{+U},\downarrow}$

$$\text{Total: } \Delta_i = \Delta_i^{\text{XC}} + \Delta_i^{\text{+U}}$$

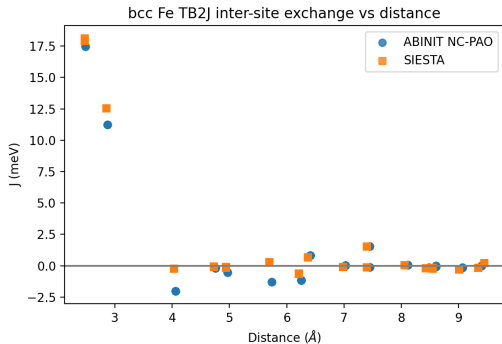
Workflow

1. ABINIT ground-state with `savetb2j 1` → NetCDF file for TB2J
2. CLI: `abinit2J.py --input <file> --elements Fe --Rmax ...`

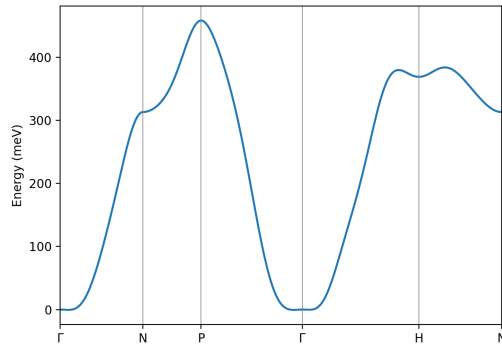
Current status

- Collinear `nsppol=2, nspinor=1`, no SOC
- Both norm-conserving and PAW pseudopotentials implemented
- PAW path is still being tested
- No merge request to ABINIT yet

Exchange parameters



Magnon band



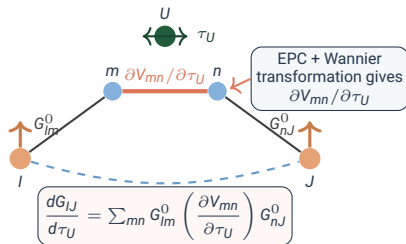
**Spin-lattice coupling parameters
by downfolding electron-phonon coupling**

Atomic displacements change hopping paths and crystal fields. Therefore the magnetic exchange parameters become functions of lattice coordinates.

$$H = H_{\text{spin}} + H_{\text{latt}} + H_{\text{SLC}}$$

$$H_{\text{SLC}} = - \sum_{I \neq J, U} O_{IJU} (\vec{S}_I \cdot \vec{S}_J) \tau_U - \frac{1}{2} \sum_{I \neq J, U, V} T_{IJUV} (\vec{S}_I \cdot \vec{S}_J) \tau_U \tau_V + \dots$$

$$O_{IJU} = - \frac{\partial^3 E}{\partial \mathbf{S}_I \partial \mathbf{S}_J \partial \tau_U} = \frac{\partial J_{IJ}}{\partial \tau_U}$$



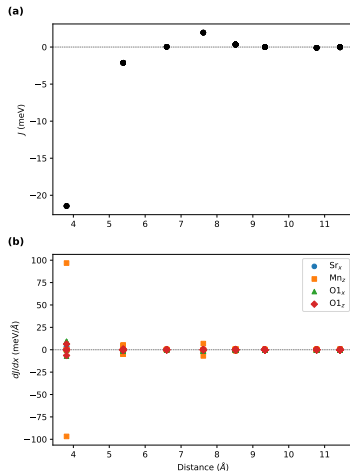
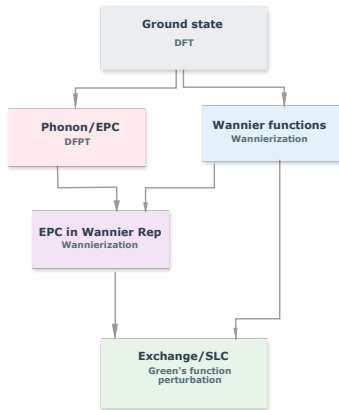
Physical picture

The unperturbed exchange uses G_{IJ}^0 . A phonon inserts $\partial V / \partial \tau_U$ on the electronic path, producing $dG_{IJ} / d\tau_U$ and therefore a displacement derivative of the LKAG exchange.

LKAG exchange and its displacement derivative

$$J_{IJ} = -\frac{1}{4\pi} \int^{E_F} d\epsilon \operatorname{Im} \operatorname{Tr}_L [\Delta_I G_{IJ}^\uparrow \Delta_J G_{JI}^\downarrow], \quad \Delta_I = H_I^\uparrow - H_I^\downarrow.$$

$$o_{IJU} = \frac{\partial J_{IJ}}{\partial \tau_U} = \frac{1}{4\pi} \int^{E_F} d\epsilon \operatorname{Im} \operatorname{Tr}_L \left[\frac{d\Delta_I}{d\tau_U} G_{IJ}^\uparrow \Delta_J G_{JI}^\downarrow + \Delta_I G_{IJ}^\uparrow \frac{d\Delta_J}{d\tau_U} G_{JI}^\downarrow \right. \\ \left. + \Delta_I \frac{dG_{IJ}^\uparrow}{d\tau_U} \Delta_J G_{JI}^\downarrow + \Delta_I G_{IJ}^\uparrow \Delta_J \frac{dG_{JI}^\downarrow}{d\tau_U} \right].$$

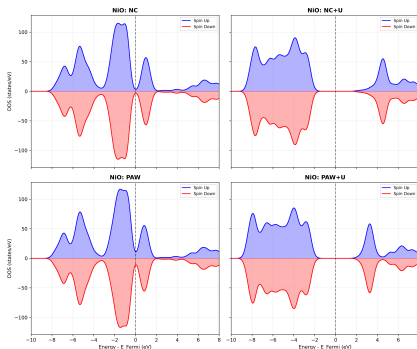




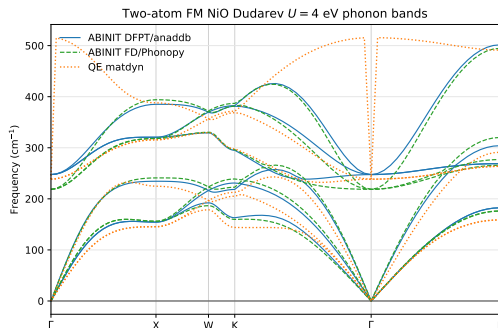
Implementation of DFT+U with norm-conserving pseudopotentials

AI-coding experiment in ABINIT

This is a part of an experiment to use AI coding within ABINIT. Rule: **not a single line of hand-crafted Fortran code.**



DOS of the ground-state AFM structure



Phonon bands for a FM structure

This talk's focus:

- Spin-lattice coupling via electron-phonon coupling downfolding
- Direct ABINIT planewave interface using pseudo-atomic orbital projectors

Outlook:

- DFT+U for norm-conserving PSP is being implemented by AI
- DFPT+U for phonons is on the way; more perturbations are coming
- Noncollinear / SOC support for ABINIT NC PAO path



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